Bayesian Hill Mixture Models for Marketing Mix Modeling: Synthetic Benchmarks, Component Resolvability, and Real-Data Evaluation

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Abstract

Marketing Mix Models (MMMs) typically use a single Hill saturation curve per channel. We study a Bayesian *predictive mixture* of K Hill curves, implemented in NumPyro with an ordering reparameterisation, post-hoc relabeling, and label-invariant convergence diagnostics. The model is a mixture over the observation likelihood rather than over latent consumer segments: at each time step the aggregate target y_t is modelled as a Gaussian mixture whose components correspond to different Hill responses on the same adstocked spend. We make no individual-level claim from this structure. All experiments—synthetic and real—use a single aggregated spend proxy $x_t = \sum_c x_{c,t}$ rather than channel-level mixtures, so the empirical scope of the paper is a controlled one-dimensional version of MMM rather than a full channel-level deployment. The question we ask is whether the extra component capacity buys better aggregate predictive density on this controlled scope and recovers interpretable response curves.

We report three experiments. (i) A controlled synthetic benchmark shows that, when the data-generating process is itself a Hill mixture, the model recovers it: estimated ELPD-LOO improves whenever the DGP has multiple Hill components and incurs only a small penalty when it does not. (ii) A *component resolvability* study varies the cosine distance between true Hill curves and identifies an empirical, model-dependent transition near $\bar{d}_{\cos} \approx 0.10$, below which posteriors tend to collapse to fewer effective components. (iii) A real-data benchmark on three Conjura organisations across three verticals (beauty & fitness, home & garden, toys & hobbies) shows that Mixture K=3 has higher estimated ELPD-LOO than single-Hill in every organisation, by +222 to +308 units; these gaps are reported subject to a paired `se_diff` that we could not recompute from the stored summaries and to Pareto-k reliability flags on Beauty & Fitness. The gain is concentrated in predictive density: holdout MAPE and CRPS do not improve uniformly, and on Beauty & Fitness they deteriorate slightly under $K = 3$.

The benchmark convergence threshold ($\widehat{R} \leq 1.05$) that 44 of 45 synthetic fits satisfy fails uniformly for Mixture K=3 on real data, where every fit reports max $\widehat{R} \geq 1.05$ and several show severe chain mixing failures (max $\widehat{R}$ up to 3.6, tail ESS per chain as low as 1–6). Re-using the resolvability axis on the real posteriors shows that this gap is organisation-dependent: Home & Garden sits at the resolvability boundary, Toys & Hobbies is consistent with three resolvable components but with chain-mixing caveats, and Beauty & Fitness shows strong multi-modality in the posterior coupled with influential observations. We report these three regimes directly rather than collapsing them into a single summary. Code, configurations, and per-fit artifacts are released for reproduction.

Keywords: Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching, component resolvability

1 Introduction

Marketing Mix Modeling (MMM) enables organisations to quantify the effectiveness of advertising spend and to optimise budget allocation. The canonical formulation fits a single response curve per channel on aggregate time-series data, combining geometric (or Weibull) adstock for carryover with a Hill function for saturation [Jin et al., 2017, Chan and Perry, 2017]. We refer to this baseline as *Standard MMM*.

Standard MMM commits to a single saturation curve per channel. There is no observational guarantee that aggregate spend–response data are well described by exactly one Hill curve: nonlinearities can arise from regime shifts, time-varying mixing weights across audiences, or simple model misspecification, and an aggregate single-Hill fit can absorb those into a biased average curve. This paper studies a Bayesian *predictive mixture* of Hill curves as a minimal capacity extension of Standard MMM, and asks two operational questions: how much component separation must be present in the data-generating process before the mixture posterior can recover it, and how well do the recovery guarantees that hold on synthetic data transfer to field data? The empirical scope is deliberately narrow: we collapse all channels into a single aggregated spend series $x_t = \sum_c x_{c,t}$ and study mixture-over-saturation behaviour on that one-dimensional problem. A channel-level mixture-of-Hill (with mixing weights and Hill components within each channel) is a direct extension but is not what we evaluate here, and we flag the cases where the single-aggregate framing materially restricts the conclusions. We deliberately do not claim that the recovered mixture components correspond to underlying consumer segments. The model is fit to aggregate daily time series and cannot identify individual-level structure; we use "component" throughout in the sense of "predictive mixture component", and discuss this scope choice explicitly in the limitations.

Contributions.

1. A controlled synthetic benchmark comparing single-Hill and Bayesian mixture-of-Hill models under known ground truth, with predictive, latent-recovery, and convergence metrics.
2. A *component resolvability* study that systematically varies the cosine distance between true Hill curves and traces the recovered Shannon effective component count as a function of true separation, identifying an empirical threshold below which the posterior collapses to fewer components.
3. A real-data benchmark on three Conjura organisations across three verticals, paired with a re-use of the resolvability axis on the real posteriors. The combination shows that mixture-of-Hill has higher estimated predictive density than single-Hill across organisations (subject to PSIS reliability and paired-SE caveats) but that publication-ready convergence is organisation-dependent in a way the resolvability framework can diagnose.

Roadmap. Section 2 situates the contribution against the MMM, Bayesian segmentation, and mixture-identification literatures. Section 3 specifies the model and the label-invariant diagnostic workflow. Section 4 consolidates the inference configuration, evaluation metrics, and publication-ready convergence criterion used throughout the paper. Sections 5–7 report the three benchmarks. Section 8 discusses what worked and what did not, Section 9 states limitations, and Section 10 describes the released artifacts.

2 Related Work

Bayesian MMM with adstock and saturation. The combination of geometric adstock with Hill saturation under a Bayesian likelihood is the canonical modern MMM formulation [Jin et al., 2017, Chan and Perry, 2017]. Subsequent open-source implementations (PyMC-Marketing, Meridian) have made this family practical at industrial scale; we keep our model in the same family so that the mixture extension is a minimal, drop-in change rather than a separate architecture. We deliberately do not extend the dynamics: Dew et al. [2024] argue that nonlinear and time-varying effects in MMM are jointly not identified from typical observational time series, so apparent saturation can be an artefact of misspecified dynamics. Their critique applies to our setting too; we do not attempt to resolve it. We instead fix the dynamics to standard geometric adstock and study what happens when only the saturation component is generalised from a single Hill curve to a Hill mixture.

Heterogeneous response and segmentation. The marketing literature on consumer heterogeneity [Wedel and Kamakura, 2000, Allenby and Rossi, 1998] works at the individual or panel level, where latent-class and hierarchical Bayes models can be identified against the data. MMM operates on aggregated daily time series and *cannot* resolve consumer-level structure from such data; in this sense the mixture-of-Hill posterior is not directly comparable to those approaches. What it offers is a strictly larger predictive model class within the standard MMM family, so that aggregate departures from a single Hill curve do not have to be silently absorbed into a biased single-curve fit.

Mixture identifiability and label switching. Finite-mixture posteriors are non-identified under permutation of component labels [Stephens, 2000, Frühwirth-Schnatter, 2006]. The label-switching problem makes per-component summaries meaningless without an additional convention, and breaks standard split-$\widehat{R}$ diagnostics [Vehtari et al., 2021]. The two standard remedies—ordering constraints and post-hoc relabeling [Papastamoulis, 2014]—trade off detailed balance against interpretability. We combine both with an additional label-invariant diagnostic (Section 3.1) and treat them as complementary rather than substitutable.

3 Model Specification

Let x_t denote aggregated marketing spend at time t and y_t the observed outcome (purchases, revenue, or a related target). Figure 1 shows the model in plate notation. The model is implemented in NumPyro [Phan et al., 2019] with JAX acceleration.

Adstock transformation. Geometric decay captures carryover from previous periods:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0, \quad \alpha \sim \text{Beta}(2, 2). \tag{1}$$

Hill saturation. Each mixture component $k \in \{1, \dots, K\}$ has its own Hill response:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}}, \tag{2}$$

with maximum effect A_k, half-saturation point λ_k, and slope $n_k \sim \text{LogNormal}(\log 1.5, 0.4)$.

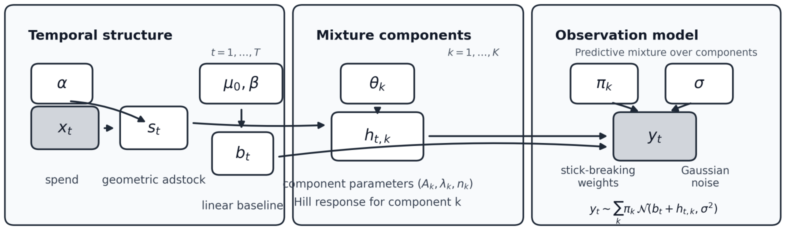

Figure 1: Graphical model. Observed variables are shaded; plates indicate replication over time steps t and mixture components k.

Mixture likelihood. The observation model is a Gaussian mixture over components:

$$y_t \sim \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(\mu_0 + \beta t + f_k(s_t),\, \sigma^2\right), \tag{3}$$

where $\pi \sim \text{Dirichlet}(\mathbf{1}_K)$ and μ_0, β capture intercept and trend.

Automatic prior scaling. To accommodate diverse data scales, priors on A_k, μ_0, and σ are computed from training-data statistics: $A_k \sim \text{LogNormal}(\log(0.3 \cdot \text{range}(y)), 0.8)$, $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$, $\sigma \sim \text{HalfNormal}(\sigma_y)$.

3.1 Identifiability and Label Switching

A finite mixture is invariant under permutations of component labels: any reordering of $(\pi_k, A_k, \lambda_k, n_k)$ yields the same likelihood. As a result, posterior summaries computed per component are meaningless without an additional convention, and standard split-$\widehat{R}$ on raw component parameters cannot distinguish genuine non-mixing from label permutations across chains [Stephens, 2000, Frühwirth-Schnatter, 2006].

Within-MCMC ordering. We impose $\lambda_1 < \lambda_2 < \cdots < \lambda_K$ via cumulative-sum reparameterization with positive increments $\delta_j \sim \text{LogNormal}(\log(s_{\max}/(K{+}1)), 0.7)$. This restricts the posterior to one canonical labelling but can distort posterior geometry near the boundary.

Post-hoc relabeling. We additionally implement unconstrained sampling followed by post-hoc reordering: each posterior draw is permuted so that $\lambda_1 < \lambda_2 < \cdots < \lambda_K$ [Stephens, 2000, Papastamoulis, 2014]. This preserves detailed balance and exposes interpretable per-component summaries.

Label-invariant diagnostics. We compute $\widehat{R}$ and effective sample sizes on three groups of quantities [Vehtari et al., 2021]:

1. *Joint log-likelihood* $\sum_t \log \sum_k \pi_k \phi(y_t \mid \mu_t^{(k)}, \sigma)$, which is label-invariant by construction.

2. *Scalar parameters* (α, σ, μ_0, β) that are not affected by label permutations.

3. *Relabeled component parameters*, after the post-hoc reordering above.

A model is treated as effectively converged when all three pass; this is the criterion we use to report a *publication-ready* fit. Concretely, a fit is publication-ready when (i) the No-U-Turn sampler produces zero divergences, zero maximum-tree-depth saturations, and a minimum chain-level BFMI of at least 0.30; and (ii) every retained mixing diagnostic satisfies $\widehat{R} \leq 1.01$ with both bulk and tail effective sample size of at least 100 per chain. The mixing diagnostics retained depend on the model: for single-Hill we use raw parameter $\widehat{R}$/ESS; for mixtures we replace the raw component-parameter diagnostics with the label-invariant diagnostics above and additionally require the relabeled component parameters to satisfy the same thresholds. These thresholds match the conservative defaults of Vehtari et al. [2021] and ensure that "pass" is a strict criterion rather than a soft one.

4 Experimental Setup

Inference. All fits use the No-U-Turn Sampler (NUTS) with two chains, init strategy `numpyro.infer.init_to_me` and diagonal mass adaptation. Warmup and sampling lengths and the target acceptance probability are model-dependent; on synthetic data we use $(600, 600)$ for single-Hill on the $K = 1$ and $K = 2$ DGPs and $(800, 800)$ on the $K = 3$ DGP, $(900, 900)$ for Mixture $K = 2$, and (1,600, 1,200) for Mixture $K = 3$. On real data the same per-model schedule is used with $(600, 600)$, $(900, 900)$, and (1,600, 1,200) respectively. Target acceptance is increased for larger mixtures to control divergences. The complete configuration is reproduced in Appendix A.

Data splits. Each time series is divided into a 75/25 train/test split. Holdout predictions inherit the adstock state accumulated during training and are evaluated at their absolute time index rather than reset to zero, so that adstock decay and the trend term are propagated forward consistently.

Evaluation metrics. We report (i) expected log pointwise predictive density estimated by Pareto-smoothed importance-sampling leave-one-out cross-validation (ELPD-LOO), with the count of $\hat{k}_i > 0.7$ observations reported alongside as a reliability flag; (ii) test-set mean absolute percentage error (MAPE), continuous ranked probability score (CRPS), and 90% predictive coverage; and (iii) for synthetic conditions, latent-mean normalised RMSE against the known DGP latent.

Publication-ready convergence. The criterion defined in Section 3.1 is applied identically across all experiments: zero divergences, zero max-tree-depth saturations, BFMI ≥ 0.30 on every chain, $\widehat{R} \leq 1.01$, and bulk/tail ESS per chain ≥ 100 on the retained mixing diagnostics (raw parameters for single-Hill; label-invariant joint log-likelihood plus scalar non-component parameters, together with relabeled component parameters, for mixtures).

Component summaries. For mixture fits we additionally report posterior mixture weights $\hat{\pi}_k$, the Shannon effective component count ${}^1D = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$ [Hill, 1973], and the mean pairwise cosine distance between active Hill response curves. The latter is computed on a fit-adaptive saturation grid that covers the active components' half-saturation range, making it independent of how the prior scale was set.

5 Synthetic Benchmark

5.1 Data-Generating Processes

We evaluate the models across three data-generating processes (DGPs): (1) *Single* ($K = 1$), a single Hill response; (2) *Mixture* $K = 2$, a two-component mixture; and (3) *Mixture* $K = 3$, a three-component mixture. Each DGP uses $T = 200$ observations and we run five random seeds per condition, so the design contains $3 \times 3 \times 5 = 45$ fits.

5.2 Results

Convergence. Under the benchmark threshold $\widehat{R} \leq 1.05$ on raw parameters for single-Hill and on label-invariant diagnostics for mixtures, 44 of 45 synthetic fits cleared the threshold (Figure 2). The single exception was the Mixture $K = 3$ model on the $K = 2$ DGP at seed 1, which reported a max $\widehat{R} \approx 1.5$ and is flagged as a benchmark failure rather than silently included. Standard single-Hill diagnostics agreed with the mixture-specific label-invariant diagnostics on every other synthetic fit. The stricter publication-ready criterion ($\widehat{R} \leq 1.01$, ESS ≥ 100 per chain, plus the HMC clauses of Section 3.1) is more demanding: 33 of 45 fits cleared it, with the remaining cases recorded as warnings (typically $\widehat{R}$ in the range $[1.01, 1.05]$). We retain the strict criterion as the headline figure for real data in Section 7, but the synthetic Section 5 plots use the looser benchmark threshold so that the comparison against published MMM benchmarks is on equal footing.

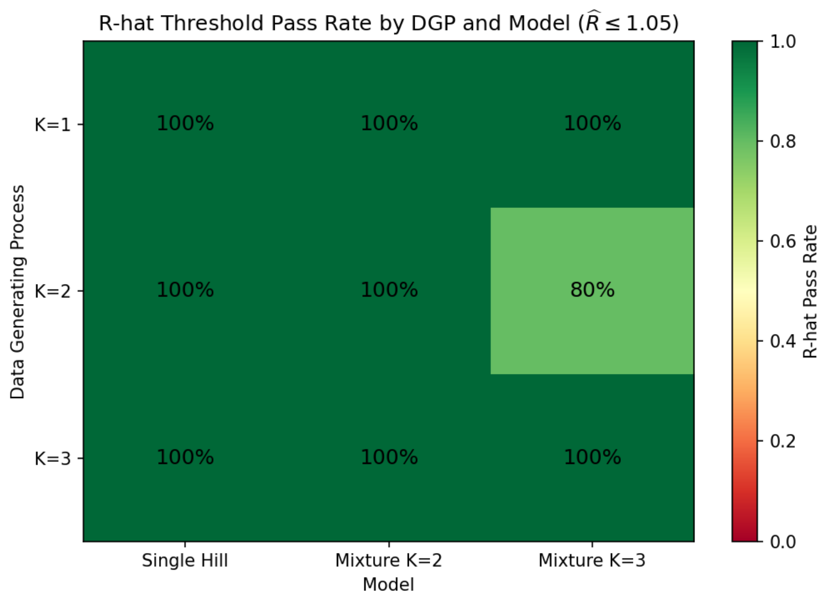

Figure 2: Benchmark convergence pass rate ($\widehat{R} \leq 1.05$ on the retained mixing diagnostics) by DGP and model on the synthetic benchmark. Eight cells reach 100%; the Mixture $K = 3$ model on the $K = 2$ DGP reaches 80% (4 of 5 seeds), with seed 1 failing at max $\widehat{R} \approx 1.5$.

Predictive performance. Figures 3 and 4 report ELPD-LOO and holdout CRPS. When the true DGP is homogeneous ($K = 1$), the three models tie within standard error. When heterogeneity is present, mixtures gain substantially in ELPD ($\Delta \approx +50$ on $K = 2$, $\Delta \approx +45$ on $K = 3$). CRPS gains are noticeably smaller than ELPD gains ($\Delta \approx 0.1$ on $K = 2$, $\Delta \approx 0.4$ on $K = 3$), indicating that the bulk of the improvement is in probabilistic density rather than in point prediction.

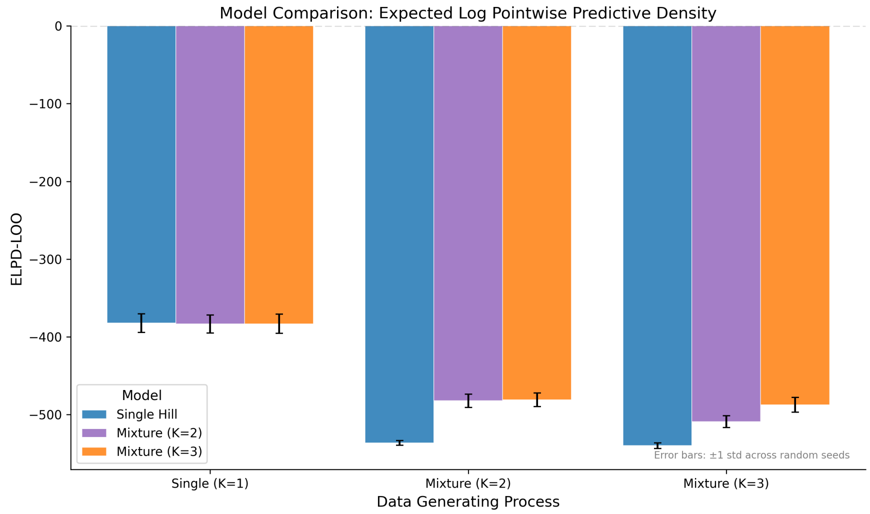

Figure 3: ELPD-LOO by DGP and model on the synthetic benchmark. Higher is better. Mixtures improve substantially when $K \geq 2$.

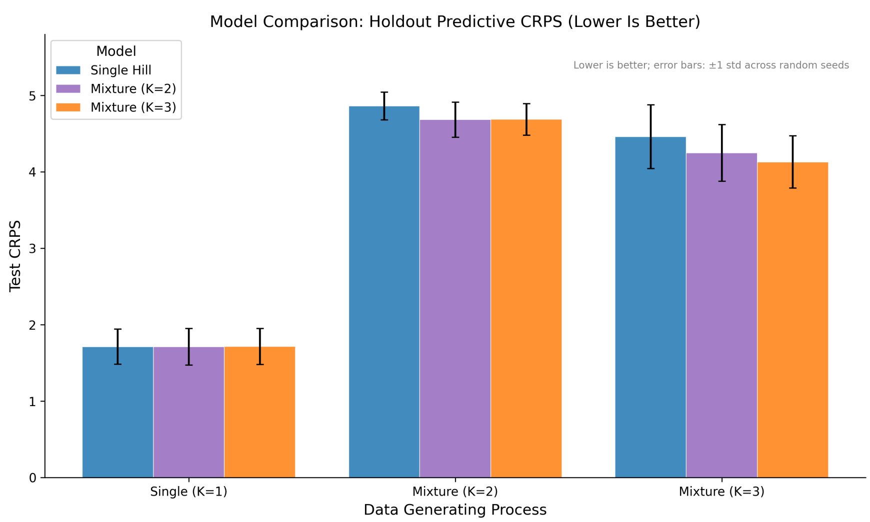

Figure 4: Holdout CRPS by DGP and model on the synthetic benchmark. Lower is better. Gains are smaller in magnitude than ELPD gains.

Latent recovery. Figure 5 shows latent-mean nRMSE on the test split. The pattern mirrors ELPD: mixtures roughly halve nRMSE under $K = 2$ ($0.14 \to 0.07$) and reduce it by a factor of ≈ 2.4 under $K = 3$ when the model is matched to the true K ($0.085 \to 0.035$). This shows that the ELPD gains correspond to genuinely better recovery of the latent response, not just better marginal calibration.

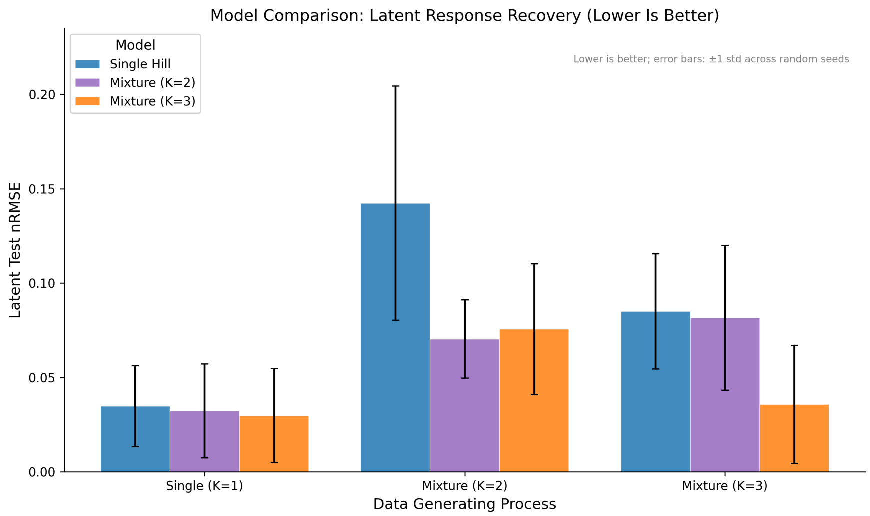

Figure 5: Latent-mean nRMSE by DGP and model. Lower is better. Mixtures recover the latent response more accurately when heterogeneity is present.

6 Component Resolvability

The previous section shows that mixtures help when heterogeneity exists, but does not tell us *how much* heterogeneity is required for the posterior to actually separate components. We address this with a controlled study.

6.1 Design

We construct nine DGP profiles by varying the spacing δ between true Hill half-points around a fixed centre c, holding amplitude and slope fixed. For $K_{\text{true}} = 1$ we use one anchor profile; for $K_{\text{true}} \in \{2, 3\}$ we use four profiles each, spanning low to extreme separation. Each profile is run with five seeds and fit with both Mixture $K = 2$ and Mixture $K = 3$, for $9 \times 2 \times 5 = 90$ fits in total. The separation axis $\bar{d}_{\cos}$ and recovery axis ${}^{1}D$ are the cosine-separation and Shannon-effective-count measures defined in Section 4; here $\bar{d}_{\cos}$ is computed on the *true* component curves and ${}^{1}D$ on the posterior mixture weights.

6.2 Results

Figure 6 plots ${}^{1}D$ against $\bar{d}_{\cos}$ across all 90 fits. Three regimes are visible:

- For $\bar{d}_{\cos} \lesssim 0.05$, both models collapse toward the trivial value: even when $K_{\text{true}} \geq 2$, the posterior cannot resolve more than ≈ 1.5–1.7 effective components.
- Between $\bar{d}_{\cos} \approx 0.05$ and 0.3, the recovered count rises sharply with separation: this is the resolvable transition region.
- For $\bar{d}_{\cos} \gtrsim 0.3$, the recovered count saturates near K_{true}. Mixture $K = 2$ saturates near 2 by construction; Mixture $K = 3$ tracks the true K and approaches 3 on $K = 3$ DGPs.

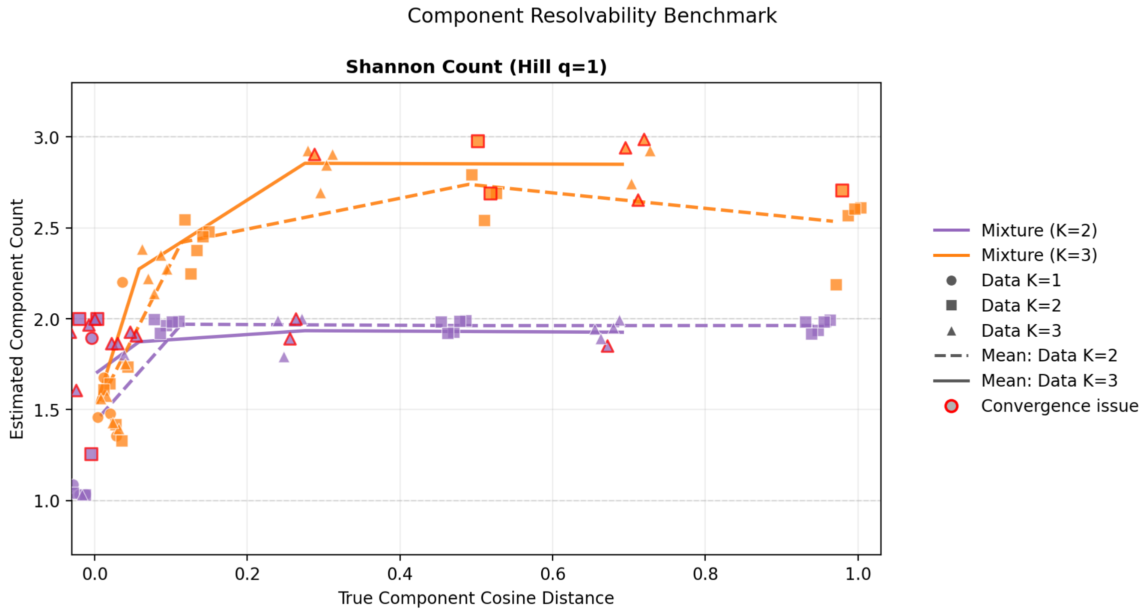

Figure 6: Component resolvability: Shannon effective count of the posterior versus mean pairwise cosine distance of the true Hill curves. Red-outlined points indicate cases that failed the publication-ready convergence criterion.

Failure modes. The publication-ready convergence criterion fails most frequently in two regimes. First, under-specified fits at low separation: Mixture $K = 2$ on $K_{\text{true}} = 3$ DGPs at low δ shows publication-fail rates of 0.6–0.8. Second, at the highest separation under $K_{\text{true}} = 3$, Mixture $K = 3$ exhibits a non-trivial failure rate (0.6 at the most extreme profile), suggesting that the strict diagnostic criterion is sensitive to posterior geometry even when the components are far apart.

Practical implication. The cosine ≈ 0.1 figure should be read as an empirical, model-dependent transition rather than a sharp threshold. The relationship between separation and recovered effective count is not monotone in K_{true} or in the fitted K: Mixture $K = 3$ on $K_{\text{true}} = 3$ recovers a Shannon count of ≈ 2.27 even at true cosine 0.058, well below the nominal 0.10 figure, and conversely Mixture $K = 3$ on $K_{\text{true}} = 3$ tv73_extreme shows publication-fail rate 0.6 at the highest separation. The picture we take away is qualitative: separations well below ≈ 0.05 are reliably under-resolved by both fitted mixtures, separations above ≈ 0.3 are reliably recovered when K matches, and the band in between is where the posterior is most sensitive to model order and seed. We use these regimes as a vocabulary for the real-data fits in Section 7.4, not as a deployable decision rule.

7 Real-Data Benchmark

7.1 Dataset and Organisation Selection

We use the public Conjura MMM dataset, which contains daily spend across nine advertising channels and several purchase-related target variables for a panel of e-commerce organisations. Following the per-organisation modelling convention of MMM practice, we treat each organisation as an independent time series and do not share parameters across organisations.

To select organisations for the benchmark, we screen all candidates with at least 200 daily observations and at least two active spend channels (non-zero on more than 10% of days), and rank them by a composite score combining series length, spend coefficient of variation, spend–target correlation, and channel breadth. We pick three top-ranked organisations from three different verticals to support a modest cross-vertical claim. The selected organisations are summarised in Table 1.

Table 1: Selected Conjura organisations for the real-data benchmark.

Vertical	T (days)	Active channels	Spend–target r	Total spend (USD M)
Toys & Hobbies	1,375	5	0.95	3.3
Beauty & Fitness	924	8	0.96	5.2
Home & Garden	1,642	5	0.80	2.1

7.2 Predictive Performance

Each organisation is fit with single-Hill, Mixture $K = 2$, and Mixture $K = 3$ under three seeds, giving 27 fits in total; inference and evaluation follow Section 4.

Table 2 reports ELPD-LOO, holdout MAPE, holdout CRPS, and holdout 90% coverage averaged across seeds. The key observation is that *Mixture $K = 3$ has the highest estimated ELPD-LOO in every organisation*, with point gaps of +222 (beauty & fitness), +308 (home & garden), and +275 (toys & hobbies) over the single-Hill baseline. Mixture $K = 2$ is intermediate. Both the magnitude and the ranking should be read alongside the paired-SE and PSIS reliability caveats below; we report them as estimates rather than as established improvements.

Table 2: Real-data predictive performance, averaged across three seeds. ELPD-LOO is reported with the per-fit PSIS-LOO standard error in parentheses (averaged across seeds); higher is better. MAPE/CRPS lower is better; 90% coverage closer to 0.9 is better. Bold marks the best ELPD-LOO per organisation.

Organisation	Model	ELPD-LOO (± SE)	Test MAPE	Test CRPS	90% cov.
Beauty & Fitness	Single Hill	$-3{,}008 \pm 86$	30.3	108	0.33
	Mixture $K = 2$	$-2{,}992 \pm 96$	29.6	101	0.37
	Mixture $K = 3$	$\mathbf{-2{,}786 \pm 48}$	34.3	122	0.54
Home & Garden	Single Hill	$-6{,}168 \pm 63$	26.0	25.0	0.87
	Mixture $K = 2$	$-5{,}985 \pm 56$	31.4	27.2	0.88
	Mixture $K = 3$	$\mathbf{-5{,}860 \pm 48}$	28.2	25.9	0.86
Toys & Hobbies	Single Hill	$-6{,}379 \pm 89$	36.4	57.3	0.94
	Mixture $K = 2$	$-6{,}346 \pm 89$	29.3	53.5	0.98
	Mixture $K = 3$	$\mathbf{-6{,}104 \pm 91}$	34.3	47.3	0.96

Two caveats apply to the magnitude of these gaps before any substantive reading. (1) The standard errors in Table 2 are per-fit PSIS-LOO standard errors of the individual ELPD point estimates, not the proper paired difference `se_diff` between two models on the same observations [Vehtari et al., 2017]. Computing `se_diff` requires pointwise log-likelihood matrices that we did not retain. As a coarse upper bound on a paired SE, treating the two models as independent gives $\sqrt{86^2 + 48^2} \approx 99$ (beauty & fitness), $\sqrt{63^2 + 48^2} \approx 79$ (home & garden), and $\sqrt{89^2 + 91^2} \approx 127$ (toys & hobbies), under which the $K = 3$ vs. single-Hill gaps are roughly 2–4 such SE in magnitude—meaningful but not as dramatic as the per-fit SE alone might suggest. The seed-level SDs in Figure 7 are Monte Carlo variability across NUTS reruns, not data-level uncertainty. (2) PSIS-LOO itself is reliable only when $\hat{k}_i \leq 0.7$ for essentially all observations. Table 3 reports the per-cell $\hat{k}_i > 0.7$ counts: Beauty & Fitness Mixture $K = 2$ has 15 such observations on average (max 21), Beauty & Fitness Mixture $K = 3$ has 3.3 (max 10), and other cells are in the single digits. The ELPD-LOO numbers for Beauty & Fitness in particular should therefore be read as ELPD estimates from a partially-degraded PSIS approximation, not as clean predictive-density measurements; the relative ordering of models is informative but the absolute magnitudes are softer.

Figure 7 visualises the same comparison. The Mixture $K = 3$ advantage is consistent in sign across organisations, while the absolute scale of ELPD-LOO varies by an order of magnitude with series length and target variance.

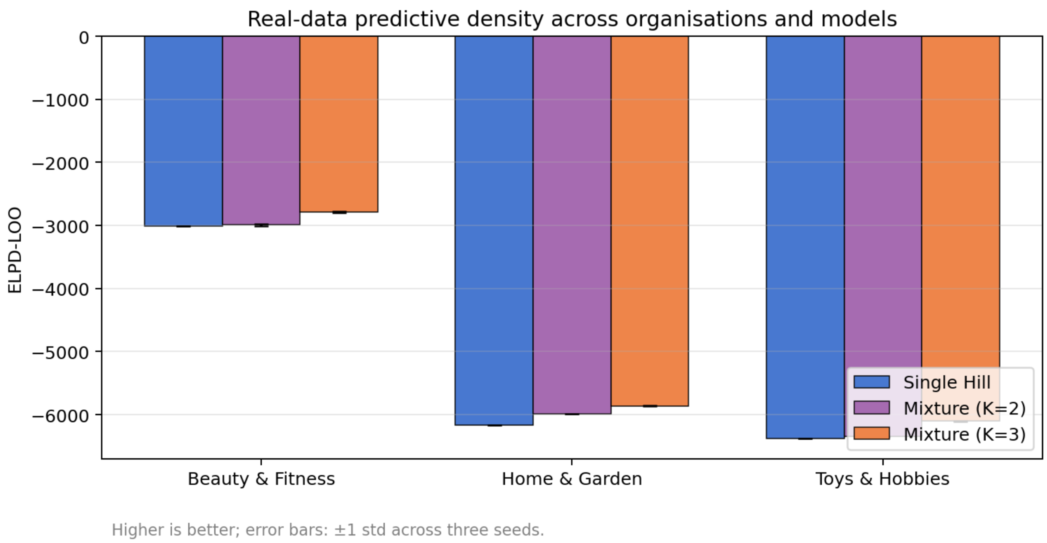

Figure 7: Real-data ELPD-LOO across the three organisations and three models. Higher is better. Error bars are ±1 standard deviation across three NUTS reruns at different random seeds and represent Monte Carlo variability of the sampler, not data-level uncertainty; in particular they should not be read as paired `se_diff` between models. Mixture $K = 3$ has the highest estimated mean ELPD-LOO in every organisation, subject to the paired-SE and PSIS reliability caveats discussed after Table 2.

The relationship between ELPD and the other metrics is heterogeneous. On *home & garden* the rank ordering of models is consistent across ELPD, CRPS, and MAPE: the mixture improves all three. On *toys & hobbies*, ELPD and CRPS both improve, while MAPE is non-monotonic. On *beauty & fitness*, ELPD improves substantially but MAPE and CRPS *deteriorate* slightly for $K = 3$, and the model remains visibly underconfident: 90% coverage rises from 0.33 (single) to 0.54 ($K = 3$) but stays well below the nominal level. The ELPD gain on this organisation therefore reflects a localised improvement in predictive density rather than a uniform predictive win; this should not be read as "$K = 3$ predicts better on beauty & fitness". We return to this in the discussion.

To make concrete what the mixture is doing, Figure 8 shows the posterior response curves for the home & garden organisation under all three models on a single representative seed. Single-Hill recovers one slowly-saturating curve. Mixture $K = 2$ resolves two components with weights 0.93 and 0.07, in which the small minority component carries a much higher saturation ceiling. Mixture $K = 3$ uses two active components with weights 0.84 and 0.14 and a third component that collapses to weight 0.02 (labelled *inactive* in the legend). The qualitative two-mode structure is robust to whether K is set to 2 or 3, which is consistent with the posterior effective component count reported below.

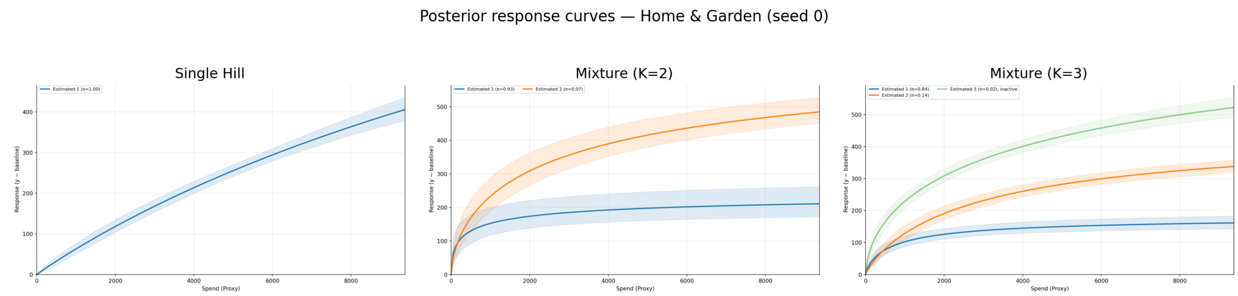

Figure 8: Posterior response curves for Home & Garden (seed 0) under Single Hill, Mixture $K = 2$, and Mixture $K = 3$. Shaded bands are 90% posterior intervals; legend entries report posterior mixture weights $\hat{\pi}$ per component. The Mixture $K = 3$ third component is labelled *inactive* because its weight collapses to 0.02.

7.3 Convergence on Real Data

Table 3 summarises the rate at which each (organisation, model) cell passes the publication-ready convergence criterion (HMC diagnostics plus label-invariant and relabeled $\widehat{R}$) across the three seeds. Two patterns are stark.

First, *Mixture* $K = 3$ *passes the publication-ready criterion in* 0/9 *fits* on real data, despite the matched-K synthetic fits passing in 5/5 runs at matched $K = 3$ (Section 5). The Mixture $K = 3$ failures on real data are also not all the same kind: Home & Garden chains hit the strict threshold only marginally, whereas Beauty & Fitness and Toys & Hobbies report max $\widehat{R}$ of 2.4–3.6 and 1.8, with tail ESS per chain of 1–6 and 23–34 respectively, and large numbers of max-tree-depth saturations. We therefore treat the three organisations' $K = 3$ fits as qualitatively different failure modes rather than a single "0/9" bucket (Section 7.4). Second, the Shannon effective count ${}^1D_{\text{post}}$ is at most 2.72 (Toys & Hobbies $K = 3$), while in two of the three organisations it stays below 1.8, suggesting that the draws rarely use the full $K = 3$ capacity in those cases.

The high Pareto-k count for the beauty & fitness Mixture $K = 2$ fit (15 observations on average) is consistent with a moderately misspecified model on data with influential outliers, and points to robustifying the likelihood as a useful next step.

7.4 Where Real Organisations Sit on the Resolvability Axis

The synthetic resolvability study (Section 6) characterises recovery as a function of *true* cosine separation between component curves. Real data has no ground truth, but we can compute the same quantity from each fit's *posterior* component summary: the mean pairwise cosine distance between the active Hill curves implied by the posterior means (A_k, λ_k, n_k), on a fit-adaptive saturation grid.

Table 3: Real-data convergence pass rates (publication-ready criterion across three seeds), Shannon effective count ${}^1D_{\text{post}} = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$ evaluated at the posterior mean of $\hat{\pi}$, active component count K_{active} (number of mixture components with posterior mean weight above $1/(2K)$ averaged across seeds), the mean and maximum number of LOO observations with $\hat{k}_i > 0.7$ across seeds, and the posterior mean pairwise cosine separation between active Hill components (Section 7.4). ${}^1D_{\text{post}}$ and K_{active} disagree because Shannon down-weights uneven mixtures: e.g. Home & Garden Mixture $K = 3$ uses two dominant components ($K_{\text{active}} = 2.00$) but with weights $\approx 0.84, 0.14$, so ${}^1D_{\text{post}} = 1.66$.

Organisation	Model	Pub. pass rate	${}^1D_{\text{post}}$	K_{active}	$\hat{k}_i > 0.7$ (mean / max)	$\bar{d}_{\cos}^{\text{post}}$
Beauty & Fitness	Single Hill	0.67	1.00	1.00	1.7 / 3	—
	Mixture $K = 2$	0.00	1.26	1.65	15.0 / 21	0.46
	Mixture $K = 3$	0.00	1.79	2.17	3.3 / 10	0.68
Home & Garden	Single Hill	0.67	1.00	1.00	0.7 / 1	—
	Mixture $K = 2$	0.33	1.27	1.94	0.7 / 1	0.09
	Mixture $K = 3$	0.00	1.66	2.00	1.7 / 3	0.12
Toys & Hobbies	Single Hill	1.00	1.00	1.00	2.7 / 4	—
	Mixture $K = 2$	0.67	1.50	2.00	1.7 / 2	0.47
	Mixture $K = 3$	0.00	2.72	2.45	3.3 / 4	0.29

Plotting the recovered Shannon effective count against this posterior separation places the real organisations onto the same plane as the synthetic resolvability fits (Figure 9).

Two important caveats apply to this overlay. First, the synthetic axis $\bar{d}_{\cos}$ is computed from the *true* component curves while the real axis is computed from *posterior-mean* curves; the latter inherits the chain-mixing problems documented in Table 3 and should be read as a point estimate with substantial uncertainty for the failed fits. Second, the $\bar{d}_{\cos} \approx 0.10$ transition in Section 6 is an empirical, model-dependent boundary rather than a sharp identifiability threshold (e.g. Mixture $K = 3$ on the $K_{\text{true}} = 3$ DGP recovers Shannon count ≈ 2.27 at true cosine 0.058).

Subject to those caveats, the picture is heterogeneous across organisations. *Home & Garden* sits at the resolvability transition: its posterior cosine separation is 0.09 under Mixture $K = 2$ and 0.12 under Mixture $K = 3$, and the Shannon count ${}^1D_{\text{post}}$ stays at 1.27 and 1.66 respectively. Among the three organisations this fit reaches the convergence threshold most closely (publication-ready 1/3 for $K = 2$, with the $K = 3$ fit failing only at the strict $\widehat{R} \leq 1.01$ bar), so its position on the diagnostic plane is the one we are most confident reporting. *Toys & Hobbies* under Mixture $K = 3$ has a posterior separation of 0.29 and a Shannon count of 2.72, which is inside the resolvable region. However, all three Mixture $K = 3$ Toys & Hobbies fits exhibit large numbers of max-tree-depth saturations and tail ESS per chain of 23–34, so the posterior-mean curves driving the 0.29 figure are themselves uncertain; we therefore present this case as consistent with three components rather than as evidence for them. *Beauty & Fitness* occupies the opposite extreme: the highest posterior separation of any organisation (0.46 for Mixture $K = 2$, 0.68 for Mixture $K = 3$), but every fit fails publication-readiness with max $\widehat{R}$ in the range 2.4–3.6 and tail ESS per chain of order 1–6 for $K = 3$. At that level the posterior summary itself is unreliable; the high posterior separation should be read as a flag for "the model finds widely-separated modes" rather than as a measurement of structure in the data.

The combined reading is that the diagnostic picture is organisation-dependent. Home & Garden lives at the resolvability boundary with reliable enough diagnostics that we can locate it on the

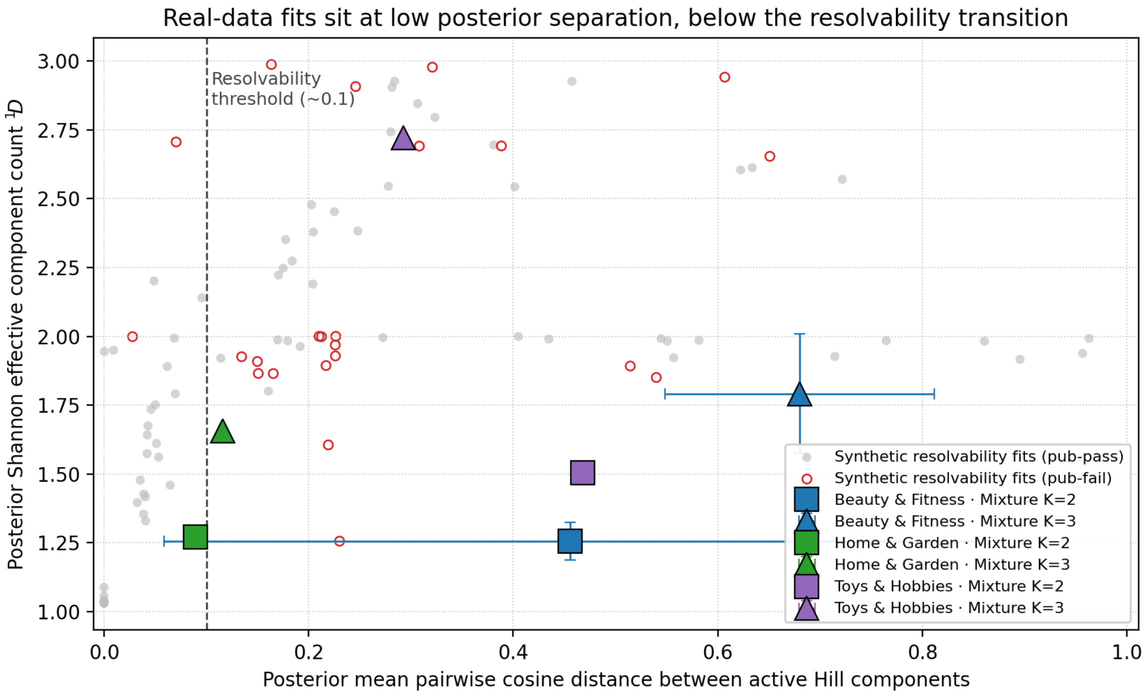

Figure 9: Posterior Shannon effective component count versus posterior mean pairwise cosine distance for the 90 synthetic resolvability fits (grey, with red-outlined fits failing the publication-ready criterion) and the three real organisations averaged across seeds (large coloured markers; error bars are ± 1 s.d. over seeds). The dashed line marks the cosine ≈ 0.10 resolvability transition identified in Section 6.

plane. Toys & Hobbies is consistent with three components subject to chain-mixing caveats. Beauty & Fitness exhibits strong multi-modality in the posterior coupled with severe chain mixing failures; substantive claims about heterogeneity there should wait for a stronger inference setup or a heavier-tailed likelihood.

8 Discussion: What Worked, What Did Not, What We Did Not Test

We organise the discussion around three honest categories so the reader can separate "the mixture is doing X" from "we did not check Y". The paper is positioned as a preprint that reports the state of an investigation rather than a finished benchmark; in that spirit, we name failures and untested directions directly.

8.1 What worked

Recovery on the matched DGP. On synthetic data where the DGP is a Hill mixture, the model recovers it. Under the benchmark threshold $\widehat{R} \leq 1.05$, 44/45 synthetic fits clear convergence (Section 5, Figure 2). Latent-mean nRMSE roughly halves under $K = 2$ DGP and is reduced by $\approx 2.4\times$ when the model is matched to a $K = 3$ DGP. ELPD gains of $+50$ ($K = 2$) and $+45$ ($K = 3$) are seen relative to single-Hill.

Predictive density on real data. Mixture $K = 3$ has the highest estimated ELPD-LOO in every one of the three real organisations, with point gaps of +222 to +308 units against single-Hill; under a coarse paired-SE upper bound (Section 7, paragraph after Table 2) these gaps are roughly 2–4 standard errors in magnitude. This result is subject to two caveats stated in Section 7: (i) the proper paired `se_diff` of Vehtari et al. [2017] could not be recomputed because pointwise log-likelihood was not retained, and (ii) Beauty & Fitness reports up to 21 LOO observations with $\hat{k}_i > 0.7$ under Mixture $K = 2$, so the PSIS approximation is partially degraded there. With those caveats in mind, the ranking of models is informative but the absolute ELPD magnitudes are softer than the raw numbers suggest.

Resolvability as a diagnostic vocabulary. The component-resolvability sweep (Section 6) gives us a concrete axis for asking "how separated must true components be for the posterior to recover the count". Re-using the same posterior-cosine quantity on real fits (Section 7.4) lets us locate the three organisations on the same plane and name their regimes (boundary / inside-resolvable / multi-modal) rather than treating real data as a single black box.

Label-invariant diagnostics. Splitting convergence into HMC clauses, label-invariant joint log-likelihood, scalar non-component parameters, and post-hoc-relabeled component parameters (Section 3.1) made the mixture posterior diagnosable in the same workflow as the single-Hill baseline.

8.2 What did not transfer

Synthetic–real convergence gap. The benchmark threshold $\widehat{R} \leq 1.05$ that 44/45 synthetic fits pass fails on 0/9 Mixture $K = 3$ fits on real data; the stricter publication-ready criterion $\widehat{R} \leq 1.01$ passes 33/45 synthetic and 0/9 real-data Mixture $K = 3$ fits. The synthetic DGPs are within the model class with stationary mixing weights, controlled Gaussian noise, and no regime shifts; the real organisations exhibit at least some of these (seasonality, channel-mix changes, outliers). We do not claim to have isolated which of those is the dominant failure cause, only that the gap is real and concentrated on Mixture $K = 3$.

Predictive density does not imply predictive accuracy. On Beauty & Fitness, ELPD-LOO improves by +222 under Mixture $K = 3$ but holdout MAPE and CRPS *deteriorate* (Section 7). The likelihood is widening around the data more than it is moving the mean toward the data. This is consistent with a Gaussian-mixture observation model picking up multi-modal residual structure (heavy tails, outliers) rather than improving point prediction. We report the ELPD gain alongside the MAPE/CRPS deterioration explicitly; we do not aggregate them.

Influential observations. The Beauty & Fitness Mixture $K = 2$ fit reports 15 LOO observations with $\hat{k}_i > 0.7$ on average; the $K = 3$ fit reports up to 10. At those levels, PSIS-LOO is itself a degraded estimator, and the ELPD gain there should be read with the Pareto-k reliability in mind, not as a clean predictive-density measurement.

Strong substantive claims from non-converging real fits. Mixture $K = 3$ on Toys & Hobbies has tail ESS per chain of 23–34 and max-tree-depth hits in the hundreds; on Beauty & Fitness, max $\widehat{R}$ reaches 2.4–3.6 and tail ESS per chain drops to 1–6. We can describe these fits and locate them on the resolvability plane, but we explicitly do not claim that their posterior-mean component summaries (e.g. "three resolvable components") are reliable measurements; they are reported as flags for further investigation.

8.3 What we did not test

These are the obvious next questions we did not attempt and have therefore not ruled in or out.

Misspecified synthetic DGPs. All synthetic DGPs in Section 5 and Section 6 are within the fitted model class (Hill family, Gaussian noise, stationary mixing weights, single aggregated channel, $T = 200$). The convergence and recovery numbers should be read as "what happens when the model is correct", not "what happens when the data look like an industrial MMM problem". We have not run DGPs with time-varying weights, heavy-tailed noise, non-Hill response shapes, or per-channel mixtures.

Alternative observation models. Two natural variations would be (i) replacing the Gaussian-mixture observation with a sum-mixture $y_t \sim \mathcal{N}(\sum_k \pi_k f_k(s_t), \sigma^2)$, which encodes "one curve composed of K Hill contributions" rather than "one of K Hill curves per observation", and (ii) replacing the Gaussian noise with Student-t to absorb the Pareto-k-flagged outliers on Beauty & Fitness. Either could change all three real-data conclusions; we did not fit them.

Stronger inference for non-converging fits. Mixture $K = 3$ on Beauty & Fitness and Toys & Hobbies might converge with longer chains, tempering, alternative initialisation, or hand-tuned mass matrices. We did not pursue this; the per-fit summaries on disk are from the configuration in Appendix A and nothing else.

Paired LOO comparisons. We report per-fit PSIS-LOO standard errors but did not retain the pointwise log-likelihood matrices needed to compute the proper paired `se_diff` of Vehtari et al. [2017]. The coarse independent-sum upper bound we report in Section 7 should not be treated as a substitute.

Causal identification. As Dew et al. [2024] argue, the joint identification of nonlinear and time-varying effects from observational time series is unresolved. Our mixture extension does not address this; it only changes the assumed saturation family. Practitioners using this model for budget decisions should triangulate against geo-experiments or incrementality tests.

9 Scope and Limitations

Section 8 discussed what we tested, what failed, and what we did not test. This section consolidates the scope choices that limit the generalisability of any positive results above.

Data scope. The real-data evaluation covers three organisations from three verticals of one e-commerce dataset (Conjura). Three organisations are enough to show that the synthetic-versus-real diagnostic gap is not a one-off, but not enough to claim a cross-organisation generalisation. We aggregate spend across channels into a single x_t to keep the model and the label-switching problem tractable; channel-level mixture-of-Hill is a direct extension but introduces label switching within each channel and we have not fit it.

Interpretation scope. The model is a predictive mixture over the observation likelihood at the aggregate daily level. It does not identify, and we do not claim that it identifies, latent consumer segments, geo segments, or any individual-level structure. Per-component summaries should be read as "which Hill response is dominant near this region of the saturation grid", not as "which audience segment is buying". Where the real-data narrative uses words like "two-mode structure" (Section 7), this refers to the shape of the response curve, not to consumer types.

Diagnostics. The publication-ready criterion is strict by design ($\widehat{R} \leq 1.01$, ESS ≥ 100 per chain, zero divergences). Real-data fits routinely fail this bar even when posterior summaries look qualitatively reasonable. We have not investigated whether a softer, application-aware criterion would distinguish "failure caused by misspecification" from "failure caused by posterior multi-modality with otherwise-good summaries". The benchmark threshold $\widehat{R} \leq 1.05$ we use for the synthetic figures is a pragmatic compromise and not a defended choice.

Priors and scales. The prior on λ depends on the empirical spend range, and the prior on A on range(y). These data-dependent priors are operationally convenient but should be made explicit in any deployment. We have not run a prior-sensitivity sweep on the multiplicative constants.

Causal identification. As above, the mixture extension does not address the joint identification problem in Dew et al. [2024]. Practitioners using this model for budget decisions should triangulate against geo-experiments or incrementality tests rather than relying on observational ELPD alone.

10 Code and Data Availability

All code, configurations, and per-case artifacts are released at `https://github.com/jpcca/marketing-mix-model` The repository contains: model definitions and inference utilities (`src/hill_mixture_mmm/`), the synthetic data generator (`data.py`), the synthetic and real-data benchmark drivers (`scripts/run_benchmark.py`, `scripts/run_real_data_validation.py`), the resolvability sweep (`scripts/run_component_resolvability_swee` and the analysis script that computes posterior cosine separation for Section 7.4 (`scripts/compute_posterior_sep` Per-fit JSON summaries containing posterior weights, label-invariant and relabeled diagnostics, and convergence flags are checked in under `paper/figures/{synthetic,real}/`. Random seeds are fixed and listed alongside the per-fit artifacts.

11 Conclusion

We presented a Bayesian mixture of Hill saturation functions for Marketing Mix Modeling, together with a label-invariant diagnostic workflow and three complementary benchmarks on a controlled, single-aggregated-spend version of MMM. Mixture-of-Hill has higher estimated ELPD-LOO than single-Hill on every one of the three real organisations, with point gaps of +222 to +308; under a coarse paired-SE upper bound these gaps are $\approx$ 2–4 standard errors in magnitude, and we report them as estimates rather than as established improvements because the proper paired `se_diff` could not be recomputed and Beauty & Fitness has degraded PSIS reliability. The gains appear in predictive density rather than uniformly across MAPE/CRPS—on Beauty & Fitness MAPE and CRPS in fact deteriorate slightly even as ELPD improves. The component resolvability study identifies an empirical cosine-separation transition near ≈ 0.10 below which the posterior tends to collapse to fewer effective components; the transition is model-dependent and should not be read as a sharp identifiability threshold. Re-using this axis on the real fits provides a diagnostic vocabulary for the three organisations: Home & Garden sits at the boundary, Toys & Hobbies is consistent with three components subject to chain-mixing caveats, and Beauty & Fitness exhibits strong multi-modality together with severe chain mixing failures. The benchmark convergence threshold that succeeds on 44/45 synthetic fits fails uniformly for Mixture $K = 3$ on real data; we interpret this gap as a directly useful finding for practitioners deploying mixture MMMs rather than a weakness to be sanded away.

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A Inference Hyperparameters

Table 4 reproduces the per-model NUTS settings used throughout the paper. Two chains are run in parallel for every fit; warmup uses NUTS dual averaging with a window size matched to the warmup length. Tree depth is capped at 10, the initialisation strategy is `numpyro.infer.init_to_median`, and mass adaptation uses a diagonal mass matrix.

Table 4: Per-model NUTS configuration. Synthetic and real-data fits differ only in warmup/sampling length; all other settings are shared. The single-Hill synthetic schedule uses 600 warmup/samples on the $K = 1$ and $K = 2$ DGPs and 800 on the $K = 3$ DGP; mixtures use the same schedule across DGPs.

Model	Warmup (synthetic / real)	Samples (synthetic / real)	Chains	Target accept
Single Hill	600–800 / 600	600–800 / 600	2	0.90
Mixture $K = 2$	900 / 900	900 / 900	2	0.95
Mixture $K = 3$	1,600 / 1,600	1,200 / 1,200	2	0.97

B Priors

Table 5 lists the priors used in every experiment. The priors on A_k and μ_0 scale automatically with the training-data range and mean to accommodate diverse target magnitudes; the prior on λ uses a cumulative-sum reparameterization (Section 3.1) with positive log-normal increments. We have not run a formal sensitivity sweep on the multiplicative constants; that is noted as a limitation in Section 9.

Table 5: Priors. Statistics with hats (e.g. $\widehat{\text{range}}(y)$, $\bar{y}$, $\hat{\sigma}_y$) are computed from the training portion only.

Parameter	Distribution	Notes
Adstock decay α	$\text{Beta}(2, 2)$	shared across components
Half-saturation increments δ_j	$\text{LogNormal}(\log(s_{\max}/(K{+}1)), 0.7)$	$\lambda_k = \lambda_1 + \sum_{j \le k} \delta_j$
Hill slope n_k	$\text{LogNormal}(\log 1.5, 0.4)$	per component
Component amplitude A_k	$\text{LogNormal}(\log(0.3 \cdot \widehat{\text{range}}(y)), 0.8)$	per component
Mixture weights π	$\text{Dirichlet}(\mathbf{1}_K)$	symmetric
Intercept μ_0	$\mathcal{N}(\bar{y}, 2\hat{\sigma}_y)$	data-dependent
Trend β	$\mathcal{N}(0, 0.5)$	on standardised t
Observation noise σ	$\text{HalfNormal}(\hat{\sigma}_y)$	data-dependent

C Synthetic DGP Parameters

Table 6 summarises the parameter ranges used to generate synthetic data for the three DGPs in Section 5 and the nine profiles used in Section 6. Per-component values are drawn uniformly from the listed ranges and held fixed across the five seeds within each profile.

Table 6: Synthetic DGP parameter ranges. The resolvability profiles fix A_k and n_k and vary only the spacing δ of the half-saturation points λ_k around a fixed centre c.

Parameter	Single ($K = 1$)	Mixture $K = 2$	Mixture $K = 3$
α	0.5	0.5	0.5
λ_k	$[1.0, 2.5]$	$[0.8, 2.8]$	$[0.7, 3.0]$
n_k	$[1.0, 2.0]$	$[1.0, 2.0]$	$[1.0, 2.0]$
A_k	$[80, 120]$	$[80, 130]$	$[80, 140]$
π_k	1	$\{0.6, 0.4\}$	$\{0.5, 0.3, 0.2\}$
σ	5.0	5.0	5.0
T	200	200	200
Train ratio	0.75	0.75	0.75